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EN.695.721.8VL

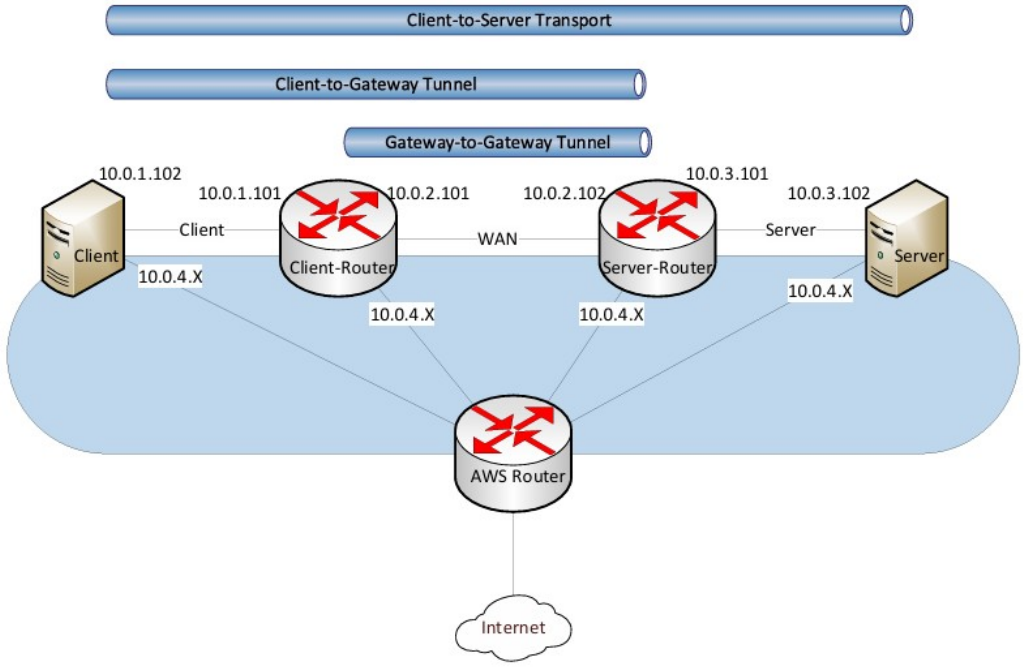
Virtual Lab: VPN with IPsec for Layer 3 Security

The goal of this project is to design a virtual network in AWS and secure Layer 3 communication using the IPsec framework. I first created an Amazon Virtual Private Cloud (VPC) and then built four subnets inside it: Client, WAN, Server, and Internet Subnet.

- Client: subnet for the client host.
- WAN: subnet for the two routers (client-router and server-router), which act as gateways between the client, the server, and the public-facing network.
- Server: subnet for the server host.
- Internet Subnet: subnet for the public-facing router.

Because I manually created the VPC using the “VPC only” option in AWS, I also had to attach an Internet Gateway (IGW) to the VPC and configure a public route table for the Internet Subnet.

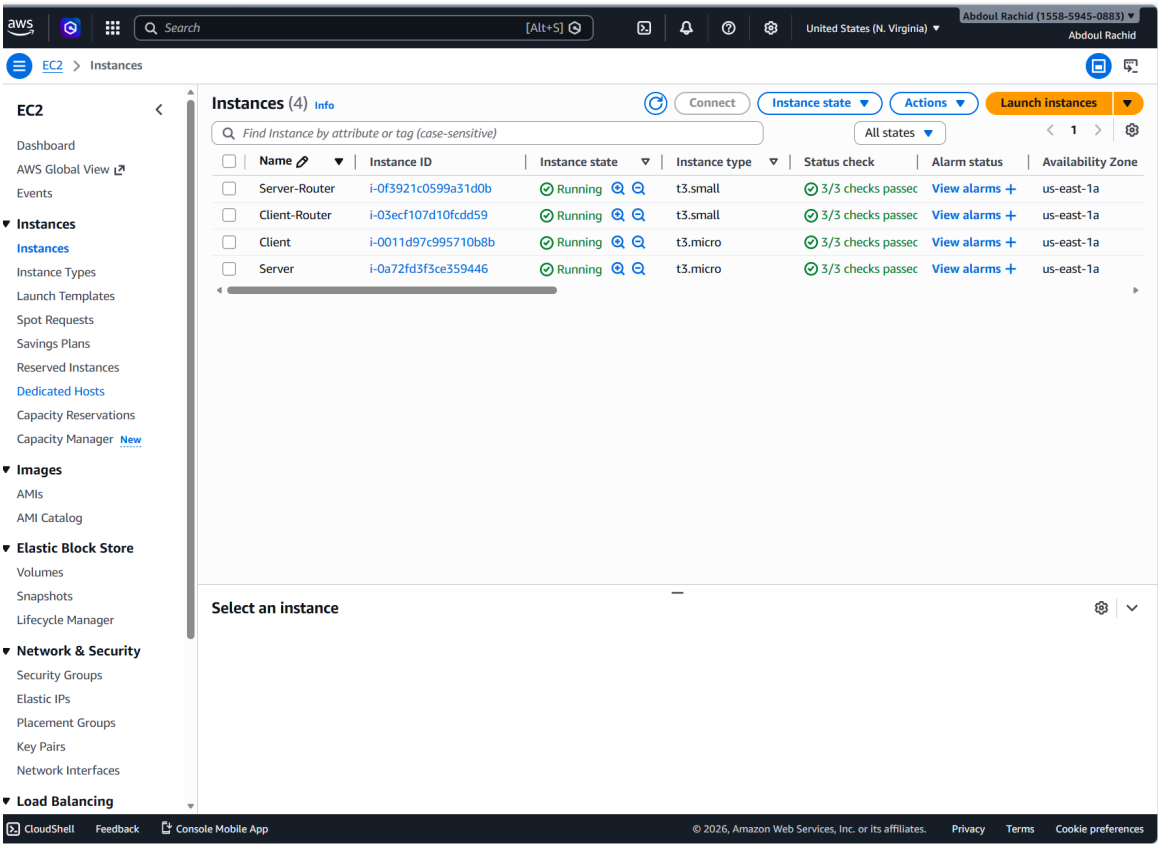
Network Diagram

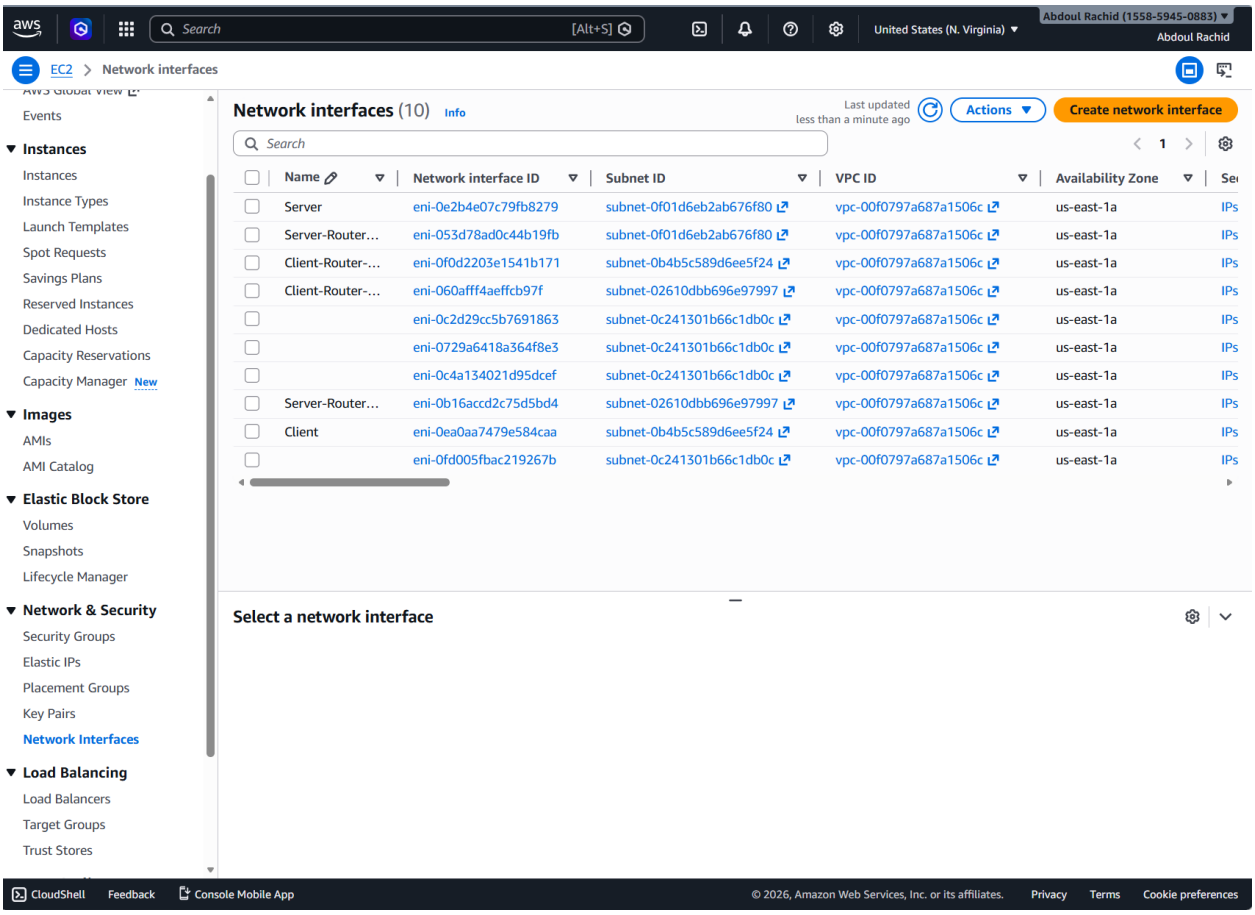


The next step was to launch the EC2 instances and attach the correct network interfaces so the implementation matched the network topology shown above.

I used t3.micro instances for the endpoints (client and server) and t3.small instances for the routers (client-router and server-router). All instances used the Amazon Linux kernel AMI.

After creating the instances, I attached each one to the appropriate network interfaces. I also disabled source/destination checking on the routers so they could forward traffic. The following images show the instances and their associated elastic network interfaces (ENIs).

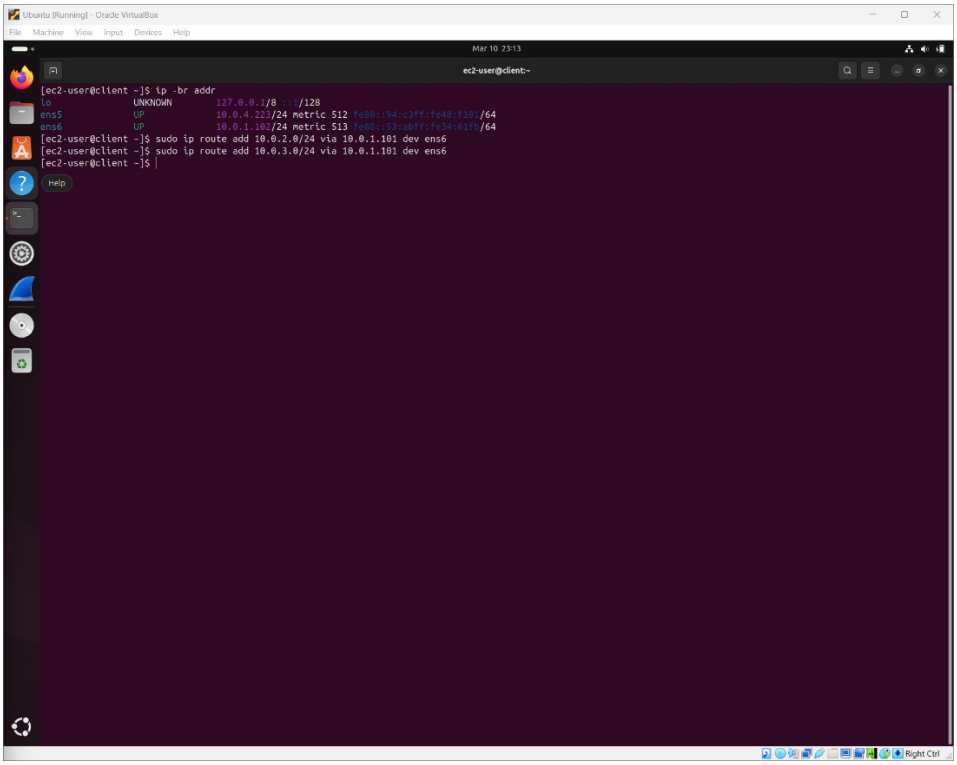




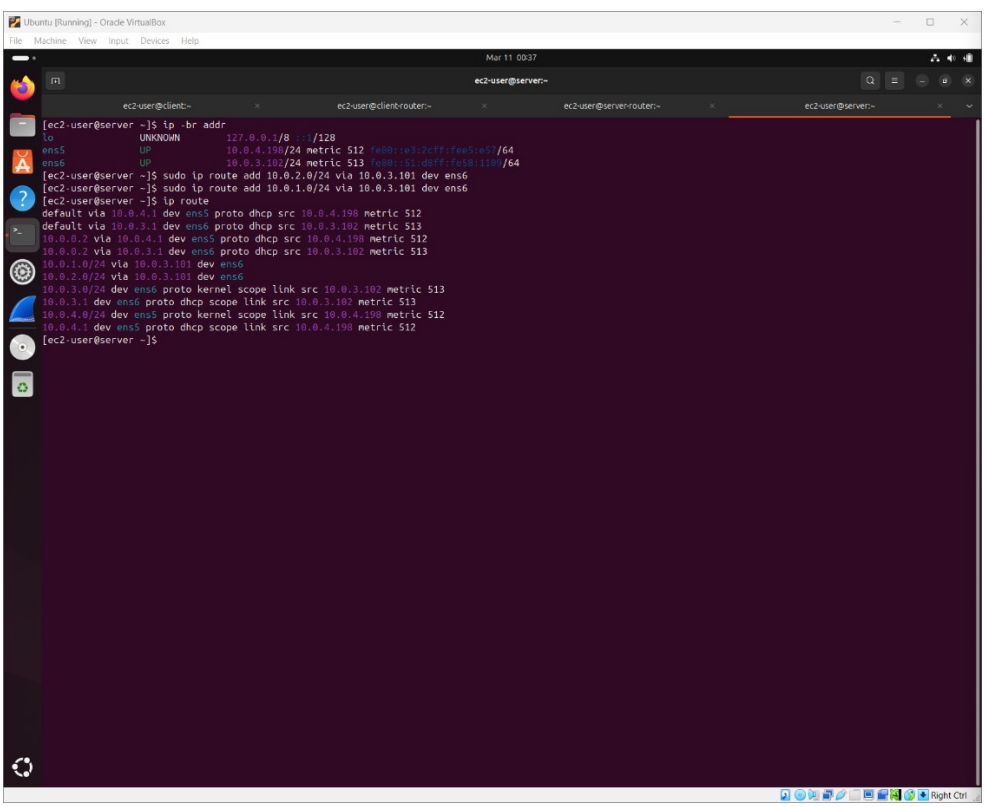
Routing Configuration

In this phase, I configured the required routes for each host and enabled IP forwarding on both routers.

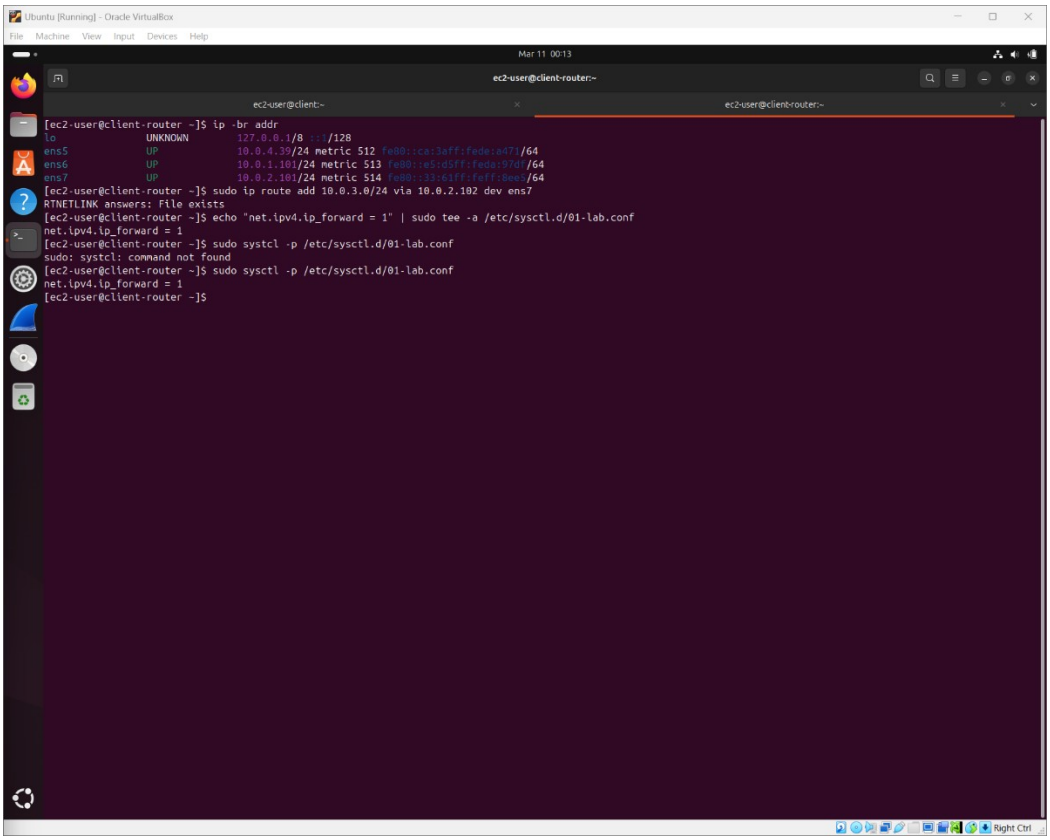
- Client instance



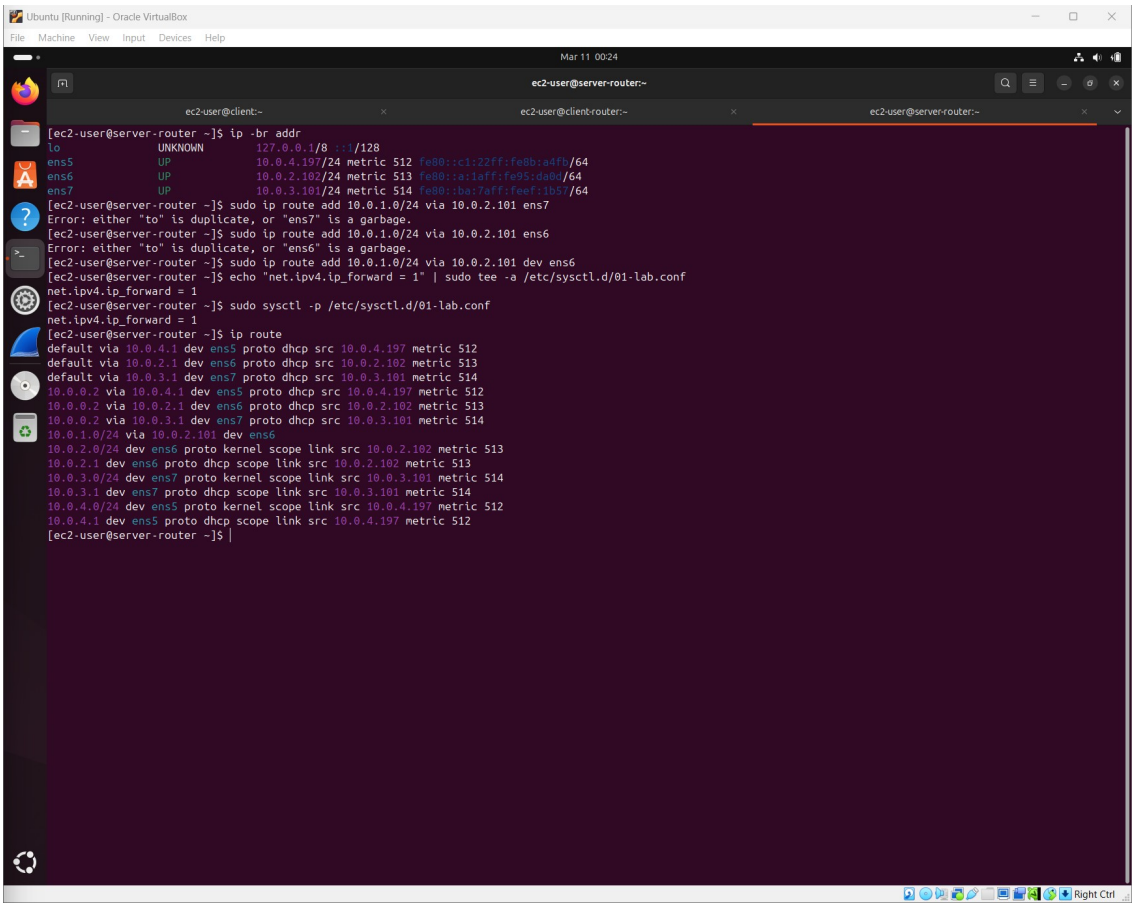
- Server instance



- Client-router instance



- Server-router instance



IPsec Configuration for Three Channels

- 1. Client-to-Server Transport Mode: 10.0.1.102 ↔️ 10.0.3.102

In transport mode, the IPsec transform is applied directly to the original IP packet being protected.

I first installed Libreswan and enabled the IPsec service on both hosts. Libreswan acts as the control plane that activates and manages IPsec inside the Linux kernel.

sudo dnf install libreswan -y

sudo systemctl enable ipsec

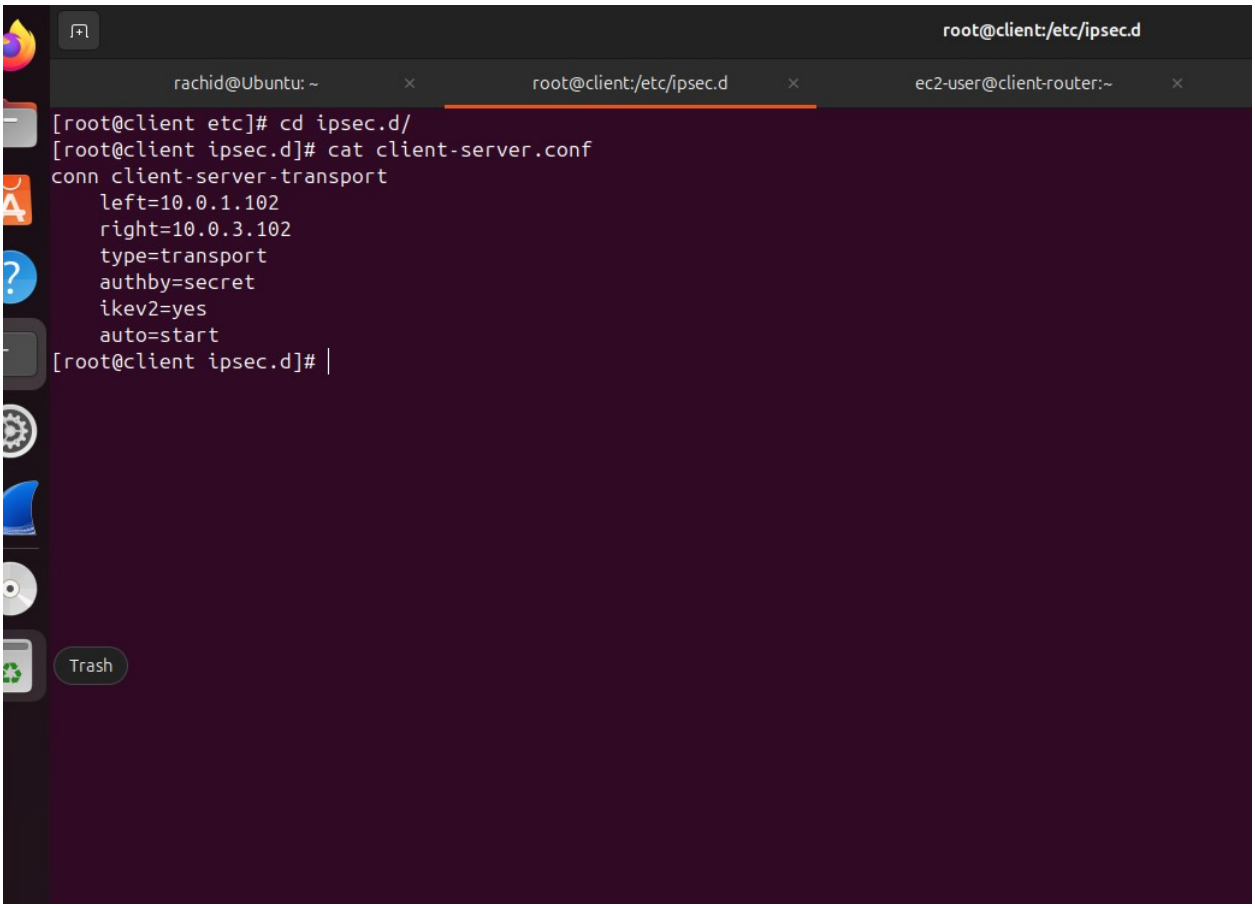
sudo systemctl start ipsec

For this configuration, I used a pre-shared key (PSK) for peer authentication. I then created the connection file that defines the security parameters for IPsec.

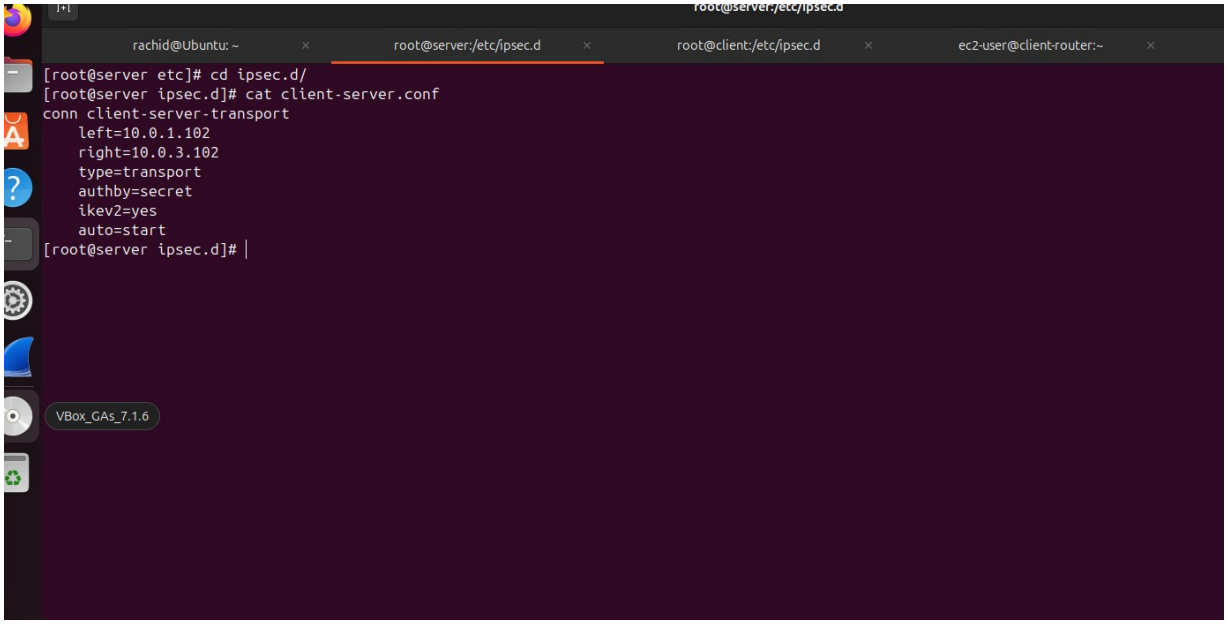
The main IPsec elements used here are transport and tunnel modes, the IKE key-management protocol, security associations, and the ESP security transform. In /etc/ipsec.d/client-server.conf, I defined the parameters for this client-to-server channel.

On each host, I configured the communicating peers as left and right endpoints. I set the mode to transport, used authby=secret for PSK-based authentication, and selected IKEv2 for key management.

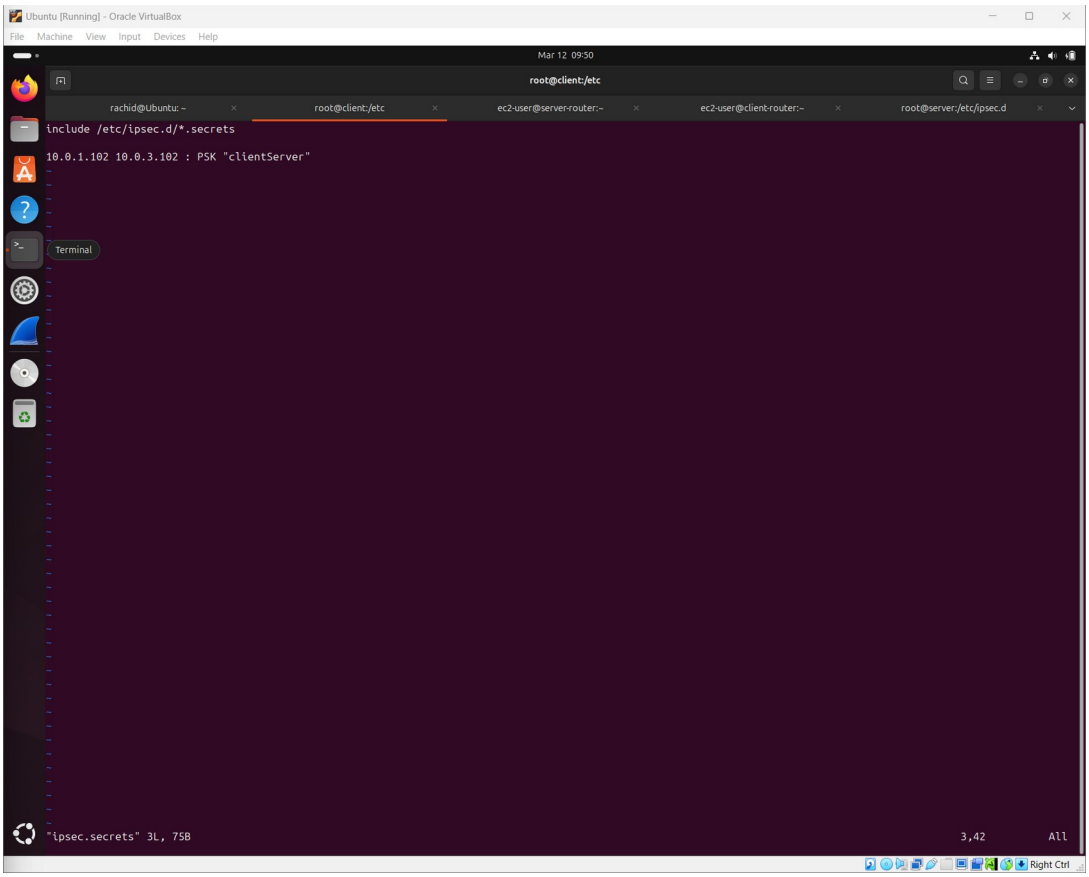
- Client side



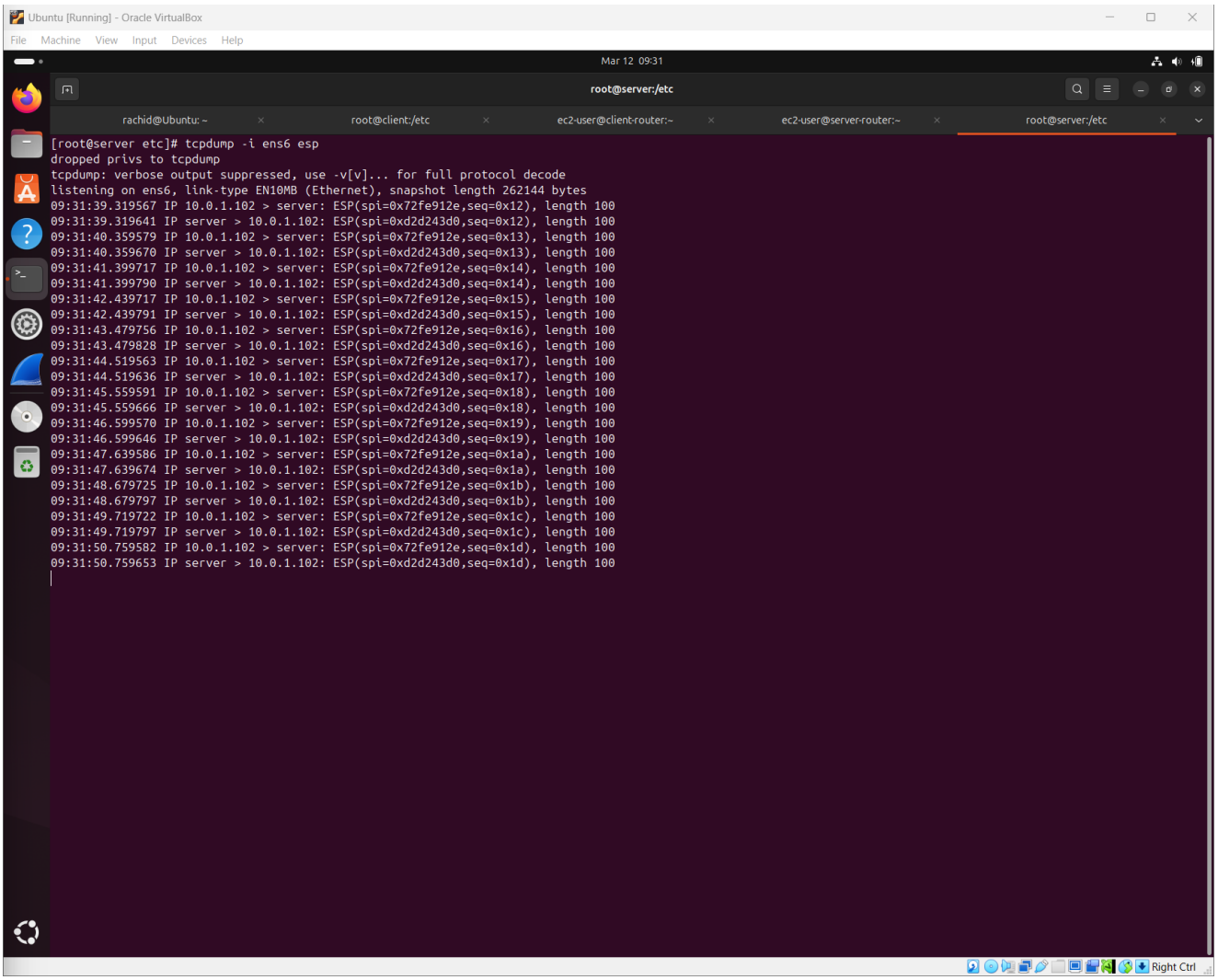
- ## Server side



Next, I configured the PSK in /etc/ipsec.secrets on both hosts.



After restarting IPsec, I pinged the server from the client and verified that ESP traffic was being generated.

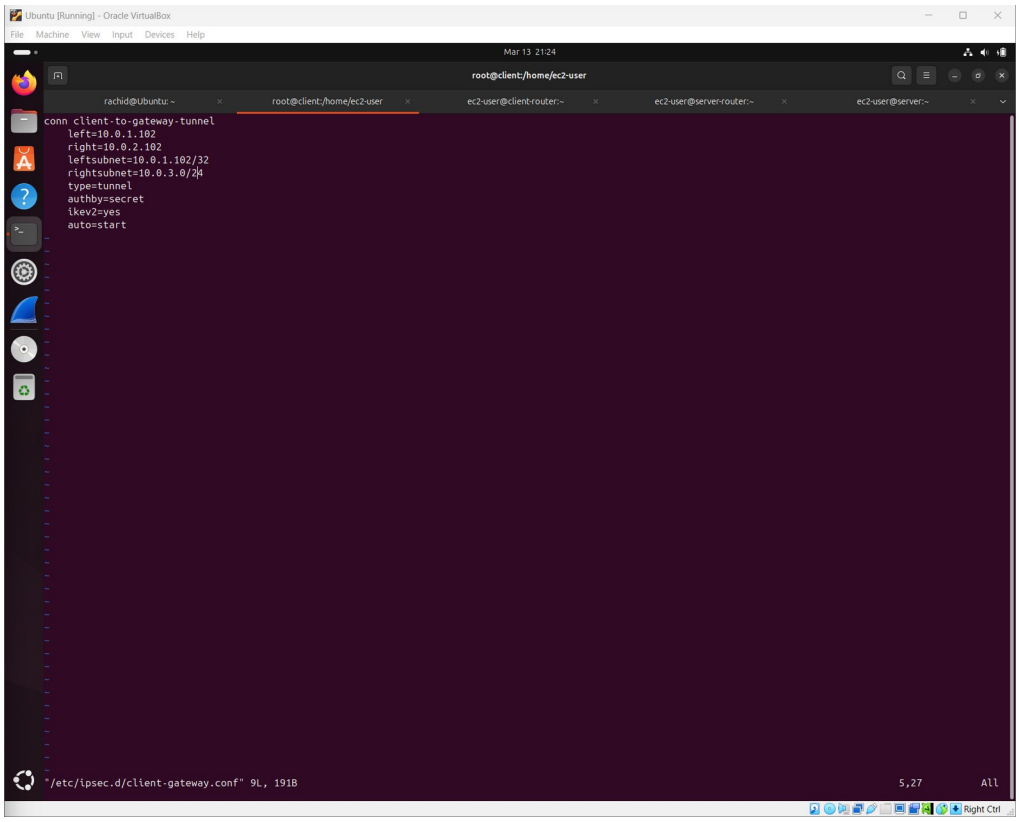


- ## 2. Client-to-Server-Router (Gateway) Tunnel Mode: 10.0.1.102 ↔️ 10.0.2.102

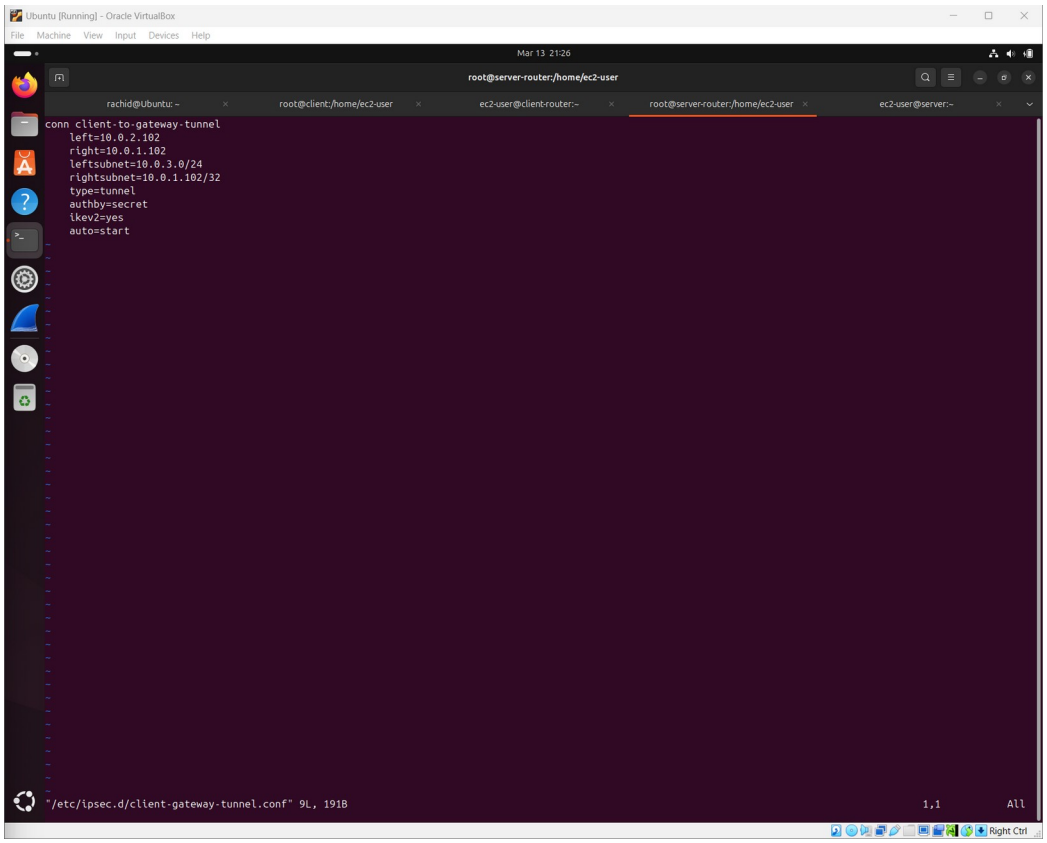
In tunnel mode, the original IP packet is encapsulated inside a new outer IP packet. The inner packet is then protected by the IPsec transform, creating a VPN channel between the two peers.

I then configured the connection file in /etc/ipsec.d.

- ### Client side



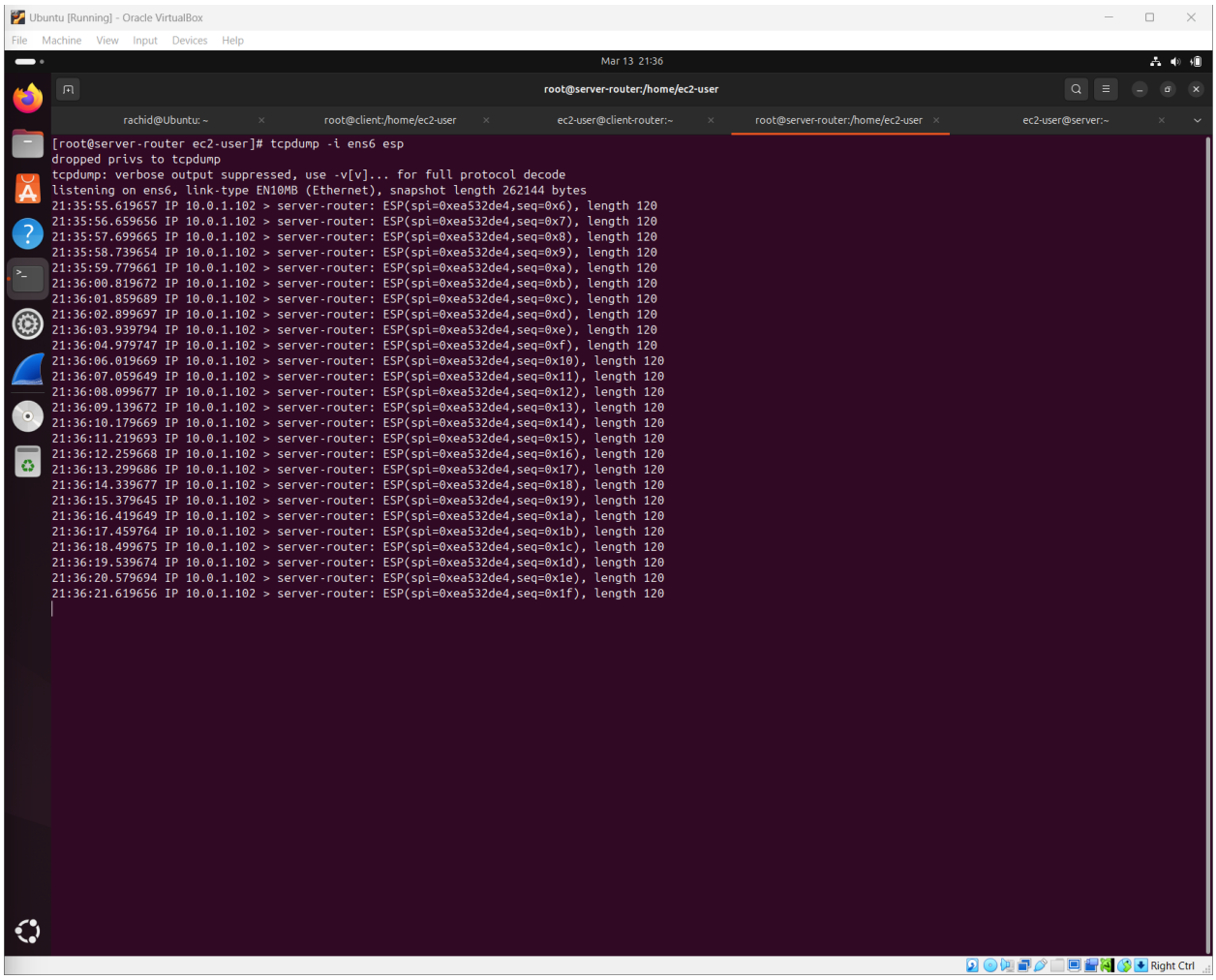
- Gateway side



The server-router serves as the entry point into the server subnet (10.0.3.0/24).

Once the tunnel was established with IPsec, I tested it by pinging the server from the client (10.0.1.102) and observing the traffic on the server-router.

```
ping 10.0.3.102
```



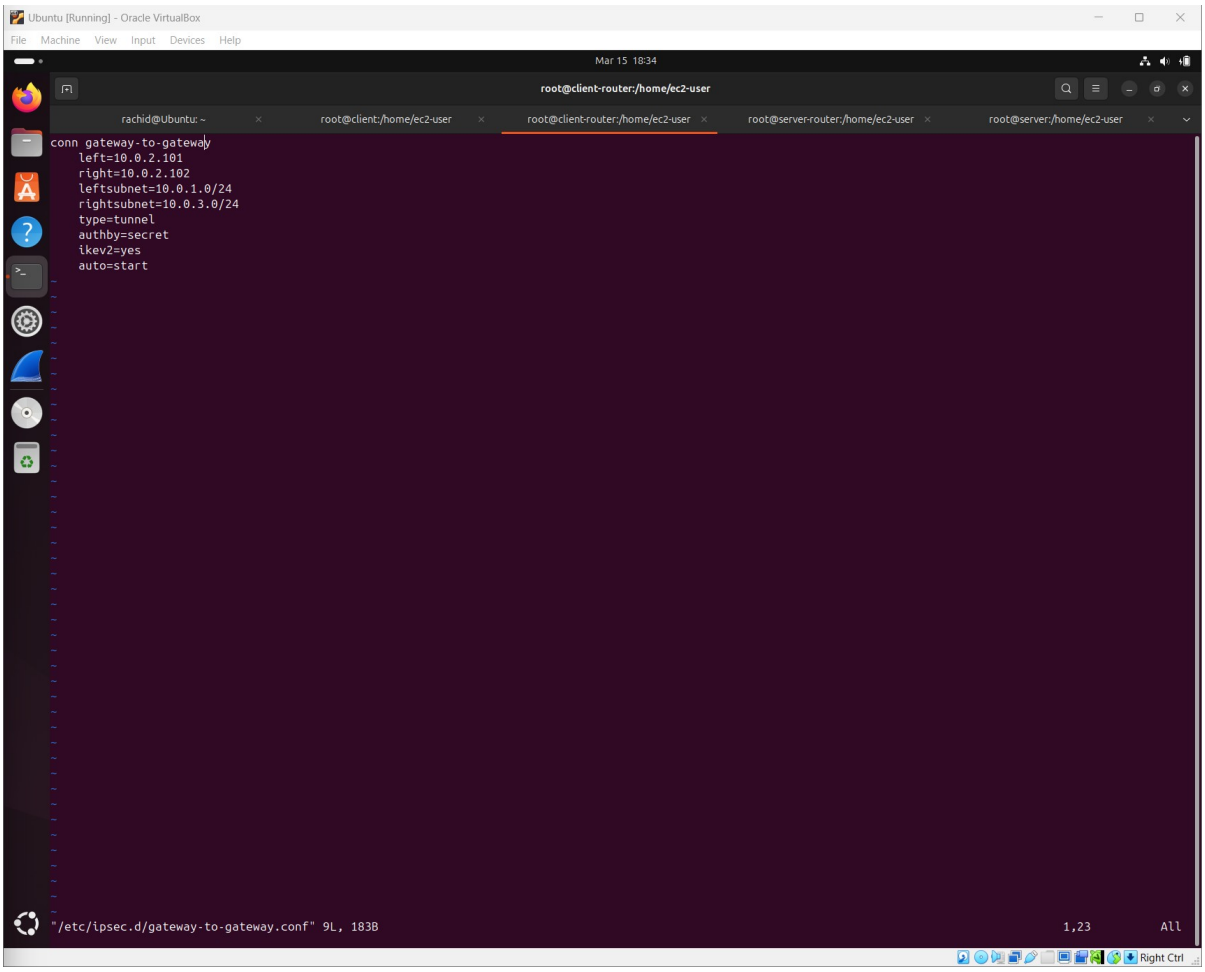
In the image above, the tcpdump capture on the server-router WAN interface shows ESP packets addressed from the client to the server-router. This confirms that the original client-to-server packet was encapsulated inside a tunnel-mode IPsec packet and that the gateway functioned as the tunnel endpoint.

3. Gateway-to-Gateway Tunnel

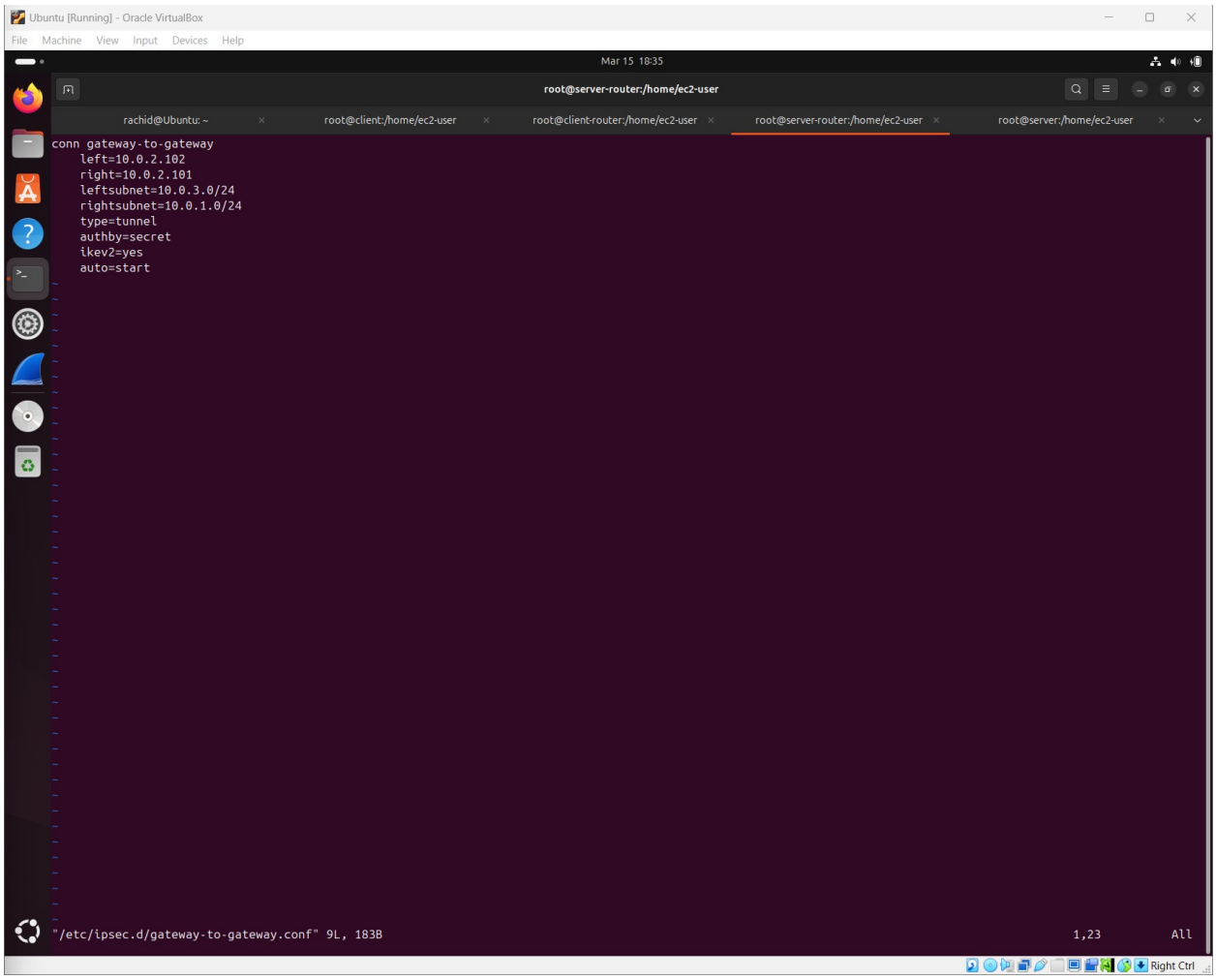
In the final stage, I built a VPN tunnel from the client-router to the server-router.

I first configured the connection file on both routers.

- Client-router side



- Server-router side



The tcpdump captures below show encrypted traffic carried with ESP on both routers, confirming the gateway-to-gateway tunnel is active.

Ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

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root@client-router:/home/ec2-user

rachid@Ubuntu: ~ | root@client:/home/ec2-user | root@client-router:/home/ec2-user | root@server-router:/home/ec2-user | root@server:/home/ec2-user

```
[root@client-router ec2-user]# tcpdump -ni ens7 esp
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens7, link-type EN10MB (Ethernet), snapshot length 262144 bytes
18:29:37.779654 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x198), length 120
18:29:38.819643 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x199), length 120
18:29:39.859661 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x19a), length 120
18:29:40.899678 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x19b), length 120
18:29:41.939677 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x19c), length 120
18:29:42.979701 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x19d), length 120
18:29:44.019658 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x19e), length 120
18:29:45.059648 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x19f), length 120
18:29:46.099642 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a0), length 120
18:29:47.139647 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a1), length 120
18:29:48.179662 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a2), length 120
18:29:49.219639 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a3), length 120
18:29:50.259653 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a4), length 120
18:29:51.299633 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a5), length 120
18:29:52.339682 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a6), length 120
18:29:53.379659 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a7), length 120
18:29:54.419677 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a8), length 120
18:29:55.459754 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a9), length 120
18:29:56.499785 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1aa), length 120
18:29:57.539660 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ab), length 120
18:29:58.579640 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ac), length 120
18:29:59.619661 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ad), length 120
18:30:00.659786 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ae), length 120
18:30:01.699691 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1af), length 120
18:30:02.739688 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b0), length 120
18:30:03.779733 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b1), length 120
18:30:04.819815 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b2), length 120
18:30:05.859674 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b3), length 120
18:30:06.899653 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b4), length 120
18:30:07.939648 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b5), length 120
18:30:08.979683 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b6), length 120
18:30:10.019682 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b7), length 120
18:30:11.059673 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b8), length 120
18:30:12.099666 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b9), length 120
18:30:13.139660 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ba), length 120
18:30:14.179665 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1bb), length 120
18:30:15.219673 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1bc), length 120
18:30:16.259667 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1bd), length 120
18:30:17.299675 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1be), length 120
18:30:18.339652 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1bf), length 120
18:30:19.379658 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c0), length 120
18:30:20.419671 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c1), length 120
18:30:21.459786 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c2), length 120
18:30:22.499792 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c3), length 120
18:30:23.539669 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c4), length 120
```

Right Ctrl

Ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Mar 15 18:30

root@server-router:/home/ec2-user

rachid@Ubuntu: ~ | root@client:/home/ec2-user | root@client-router:/home/ec2-user | root@server-router:/home/ec2-user | root@server:/home/ec2-user

```
[root@server-router ec2-user]# ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
ens5             UP             10.0.4.197/24 metric 512 fe80::c1:22ff:fe8b:a4fb/64
ens6             UP             10.0.2.102/24 metric 513 fe80::a:1aff:fe95:da0d/64
ens7             UP             10.0.3.101/24 metric 514 fe80::ba:7aff:feef:1b57/64
ip_vti0@NONE     DOWN
[root@server-router ec2-user]# tcpdump -ni ens6 esp
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens6, link-type EN10MB (Ethernet), snapshot length 262144 bytes
18:29:50.259787 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a4), length 120
18:29:51.299756 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a5), length 120
18:29:52.339803 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a6), length 120
18:29:53.379775 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a7), length 120
18:29:54.419804 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a8), length 120
18:29:55.459872 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1a9), length 120
18:29:56.499922 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1aa), length 120
18:29:57.539790 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ab), length 120
18:29:58.579795 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ac), length 120
18:29:59.619790 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ad), length 120
18:30:00.659910 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ae), length 120
18:30:01.699833 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1af), length 120
18:30:02.739835 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b0), length 120
18:30:03.780267 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b1), length 120
18:30:04.819933 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b2), length 120
18:30:05.859799 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b3), length 120
18:30:06.899777 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b4), length 120
18:30:07.939769 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b5), length 120
18:30:08.979834 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b6), length 120
18:30:10.019803 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b7), length 120
18:30:11.059792 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b8), length 120
18:30:12.099790 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1b9), length 120
18:30:13.139787 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1ba), length 120
18:30:14.179788 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1bb), length 120
18:30:15.219788 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1bc), length 120
18:30:16.259798 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1bd), length 120
18:30:17.299797 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1be), length 120
18:30:18.339774 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1bf), length 120
18:30:19.379786 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c0), length 120
18:30:20.419810 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c1), length 120
18:30:21.459907 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c2), length 120
18:30:22.499915 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c3), length 120
18:30:23.539796 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c4), length 120
18:30:24.579772 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c5), length 120
18:30:25.619799 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c6), length 120
18:30:26.659794 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c7), length 120
18:30:27.699768 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c8), length 120
18:30:28.739772 IP 10.0.2.101 > 10.0.2.102: ESP(spi=0xd4a955a9,seq=0x1c9), length 120
```

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